Name:
ID:

CSE1100 – Introduction to Computer Engineering
Quiz 2 - 01/11/2019

1) Convert the following decimal number into its IEEE floating point representation:

-18.25

18.25 → $18\frac{1}{4}$ → 10010.01 → $1.001001 * 2^4$

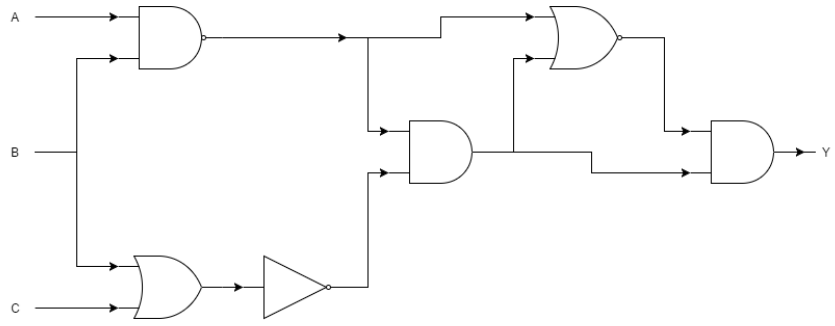
Sign: 1

Exponent: 10000011 (127+4=131)

Fraction: 00100100000000000000000

1 10000011 00100100000000000000000

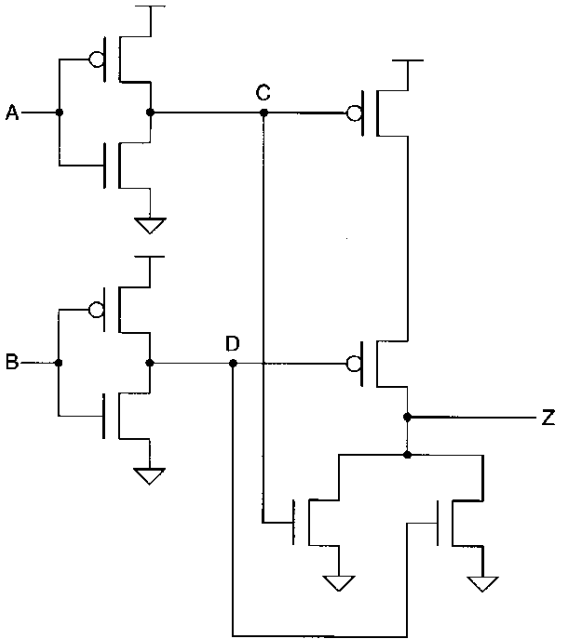
2) Given the logic circuit below, fill in the truth table for the output value Y.



Y = ((A . B)' + ((A . B)' . (B + C)'))' . ((A . B)' . ((B + C)'))

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

3) For the transistor-level circuit in the given figure, fill in the truth table. What is Z in terms of A and B (write the logical equation)?



A	B	C	D	Z
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

Z = A AND B

4) Compute the following. Write your results in binary.

a. 01010111 **AND** 11010111
   01010111
b. 10101 **OR** 11001
   11101
c. **NOT** (**NOT** (101) **AND** (110))
   101

5) Following is a 2-to-4 decoder with inputs A and B. Design a 3-to-8 decoder with inputs A, B, and C.

